

Quick look: LTE Release 10 vs 802.16m WiMAX

1. **OFDM.** Both LTE access and WiMAX access are OFDM-based technologies and they operate at a frequency band different from current 2G/3G systems.
2. **IP-based flat architecture.** The core network architecture for both is flat IP-based. The cores are converging onto a single platform, and some vendors are already providing single cores to handle 2G/3G and 4G (BWA) systems.
3. **Voice and SMS.** Both WiMAX and LTE are IP networks and do not support voice as a native capability. There was a proposal that LTE could make use of the voice capability of 2G/3G networks with VoLGA. But that proposal has been rejected by the International Telecommunications Union (ITU) and voice is only supported using an IP multimedia subsystem (IMS). However, in comparison with WiMAX, LTE is expected to be better in terms of voice and SMS support.
4. **Interoperability.** LTE is designed to co-exist with GSM, WCDMA and HSPA networks; just that the RAN needs to be upgraded in a phased manner (just like the recent 3G roll out was possible through an upgrade of existing GSM/EDGE networks in a phased manner). The non-3GPP WiMAX is not interoperable with GSM and WCDMA networks, yet.
5. **Backward-compatibility.** Both LTE as well as WiMAX are not backward-compatible with earlier 3G standards. However, LTE-Advanced will be backward compatible with LTE; and WiMAX 802.16m will be backward-compatible with 802.16e.
6. **Transition.** The ease of transition from an existing technology to a new one is a key consideration when moving from one generation to another. In cases where a service provider has an existing network with a large number of subscribers, a phased transition is expected. In such a case, the cost of upgrading the core network to support both old and new subscribers deserves consideration. Therefore, in Greenfield projects like rural broadband access, or private networks (for defence, public safety, smart grid, etc), we see WiMAX as a good option. However, for telecom operators covering dense deployments and having an existing 2.5G/3G network, LTE may be a better choice.

WiMAX is more data-centric for fixed and nomadic applications, LTE is more focused on mobility and wide area coverage. LTE has a legacy of design and deployment experience in the mobility arena since it comes from the mature 3GPP family of technologies.

"LTE is the fastest developing system in the history of mobile communication. In its first year, LTE usage soared from zero to 150 million people who have access to LTE networks today. Ericsson has supplied the large majority of these commercial LTE networks and has signed contracts with five of the top ten ranked operators by global revenue in 2010. Verizon's LTE network is the world's largest commercial installation to date. Just like the world's first LTE network, launched by TeliaSonera in Sweden, it was supplied by Ericsson," says Dhawan.

Just as WiMAX 802.16m is backward-compatible with its predecessor, LTE Release 10 is also backward-compatible with LTE Release 8. However,

both LTE and WiMAX are not backwards-compatible with older networks. "In contrast to WCDMA/HSPA, LTE uses only packet core. Also, its modulation and access being OFDM in contrast to CDMA for WCDMA/HSPA, it is a disruptive technology with respect to these earlier 3GPP standards. Hence, compatibility/migration is not possible," explains Dr Borkar.

However, one advantage that both HSPA+ and LTE have over WiMAX is their interoperability with WCDMA and GSM networks. Experts feel that the coexistence of LTE, HSPA and other 3G technologies will ensure service continuity as LTE roll-outs and subscriber uptake proceeds at different rates in different regions. Multi-standard radio equipment that can deliver LTE, WCDMA/HSPA and GSM all from a single base station are available. This gives mobile operators the freedom to balance voice and data traffic across their different networks efficiently, together with the ability to serve all subscribers, irrespec-

tive of their network or device, from the same radio base station.

TD-LTE: The latest 4G craze

The 3GPP portfolio now includes a TDD version of LTE, popularly known as TD-LTE or TDD-LTE. TD-LTE is capable of running on the same frequency band (2.6 GHz in the US) as WiMAX, which is also a TDD technology. This part of the spectrum is much cheaper and has much less traffic as compared to the one used by the regular FDD version of LTE.

TD-LTE is touted to have many advantages over the FDD one. Regular LTE networks carry two separate signals, one for data travelling in each direction, whereas TD-LTE works over a single channel, simply allocating upload or download bandwidth depending on what you do. Plus, TD-LTE is so similar to normal LTE that the same chip can be used for both, making it easy for device makers to adopt the new technology. What's more, TD-LTE is fully-compatible with existing GSM/HSPA technologies with support for seamless mobility between these technologies. It is possible to upgrade even earlier WiMAX networks to the 4G TD-LTE, unlike the incompatibility between normal LTE and WiMAX.

Not surprisingly, the 2011 episode of CommunicAsia seemed to suggest the gaining popularity of this new standard, which has a Chinese origin. Apple is also considering a TD-LTE device.

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"As per an Ericsson Consumer Insights study, out of nearly 5 billion people who are estimated to have mobile broadband by 2016, about 95 per cent will be served by HSPA, WCDMA and LTE networks," says Dhawan.

The current distribution of the various standards varies significantly across different regions and countries. A significant part of the world still uses GSM/GPRS/EDGE as the primary broadband technology, being only in the very early stages of introducing 3G